

Priyanka Chatterjee

AI Backend Engineer · 9,300★ Open-Source Contributor · Python · RAG · MCP · 3.5+ Years
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SUMMARY

AI Backend Engineer with 3.5+ years building **agentic AI** systems — **Model Context Protocol** servers, **multi-agent** LLM orchestration, and **RAG** — in **Python**, backed by production backend and data-layer work across **MongoDB**, **PostgreSQL**, **Docker**, and **AWS**. Built the Steampipe **MCP** server end to end and a **RAG**-based **Pydantic AI** agent (deployed to **AWS** Lambda with **streaming** output) at Turbot HQ — one of 120+ merged **pull requests** across its 9,300★ open-source suite.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript (ES6+), SQL, Bash, HCL

AI & Agentic Engineering: LLM Application Development, Agentic Workflows, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), OpenAI Agents SDK, Pydantic AI, Vercel AI SDK, Prompt Engineering, Claude Code, Cursor, Codex, Logfire (LLM observability)

Databases & SQL: PostgreSQL (Postgres), MongoDB, DuckDB, SQLite, Redis (ElastiCache), AWS S3 vector storage; jsonb, CTEs, query tuning, indexing, transactions

Cloud & Serverless: AWS (Lambda, EC2, RDS, S3), Azure, GCP, OCI, AliCloud

IaC & DevOps: Docker, Kubernetes (working knowledge; collaborate with platform teams), Terraform, GitHub Actions, Git

Observability: Datadog, OpenTelemetry (OTel), Logfire; metrics, logging, tracing, incident triage

Backend & APIs: Microservices, Event-Driven Architecture, Distributed Systems, Serverless, REST, gRPC, OAuth 2.0 (Auth0), JWT

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Built a **RAG**-based internal **Pydantic AI** agent in **Python** to resolve cross-team queries by ingesting the documentation and code graph for all internal repos into **AWS** S3 vector storage via **embeddings**; deployed it to **AWS** Lambda and integrated it with the Slack Bot API as the UI, resolving onboarding and cross-team queries instantly.
- Designed a **multi-agent** workflow on the OpenAI Agents SDK to orchestrate LLM agents that generate cloud preventive-policy commands, validate them in a live test environment, self-correct failures, and score accounts against an internal security metric.
- Built the Steampipe **Model Context Protocol (MCP)** server end to end in **Python**, giving LLM agents standardized, real-time access to enterprise cloud and security data for retrieval and analysis.
- Debugged a production performance bottleneck by reading the query plan for slow **PostgreSQL** compliance-audit queries — rewriting **AWS** IAM policy-evaluation queries with jsonb array elements and multi-stage CTEs, fixing the underlying indexing rather than the symptom — improving query performance 40% across 10,000+ resources.
- Diagnosed and resolved a production incident by reading pod logs and status for a Go service stuck in a **Kubernetes** CrashLoopBackOff, collaborating with the platform team to restore service.
- Designed and developed an internal REST API microservice in Go using the Gin framework as part of a greenfield build of Turbot Guardrails, a Preventive Security Posture Management (PSPM) SaaS product; secured endpoints with OAuth 2.0 (Auth0), containerized with **Docker**, and deployed to **AWS** EC2.

PROJECTS

General-Purpose AI Chatbot / Agent (Model Context Protocol) (Jun 2025 – Aug 2025)

Agentic chatbot/agent framework built with **Pydantic AI** featuring configurable external **MCP** servers via a persistent **MongoDB**-backed registry, JSON config loader with strict input validation, per-server failure isolation, native tool calling, mid-conversation multi-model switching (OpenAI, Gemini, Anthropic, Mistral), conversation-history persistence, sync + **streaming** output, prompt-injection safeguards, and Logfire observability.

EDUCATION

University of Engineering & Management (UEM) Kolkata, WB, IN

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design